

only gives us an idea about how well our model does on data used to train it which results in biased error values. If the data volume is large enough to represent the population, you may not need the validation techniques. In real-world scenarios, data samples may not truly reflect the real world scenario. This is where validation techniques come into the picture.

Here we talk about different validation techniques:

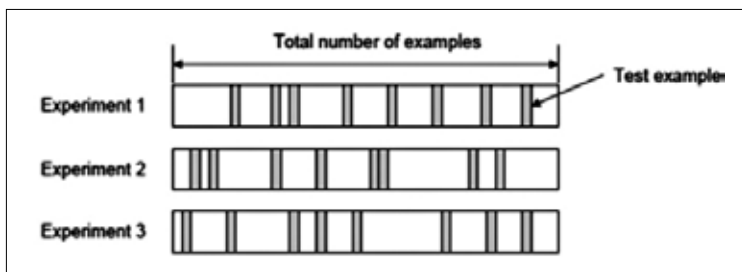
- Hold-out
- Random Subsampling
- K-fold cross-validation
- Stratified K-fold cross validation
- Leave-P-out cross validation

Holdout Method

To avoid any training error, the data is split into two datasets labeled as a training and a testing dataset. Usually, a 60/40 or 75/25 or 80/20 split is preferred. This technique is called the hold-out validation technique. We try to create the training and test dataset with equal distribution of all the different classes of data. This process is also known as stratification. It suffers from the problems of high variance. It is due to the fact that we have no idea which data points will end up in the validation set and the results vary entirely for different sets. The validation and the testing dataset are the same in this case.

Random Subsampling

Random sets of data are picked up from the original dataset and combined to form the validation set. The remaining data constitutes the training data. The figure below is an accurate representation of random subsampling. The average of the error rate of each iteration gives the error rate of the model.



Random Subsampling